

React JS Hands-on Lab

Conditional Rendering – Ticket Booking App

Objective

Create a React Application named **ticketbookingapp** where a guest user can browse a page showing the flight details, whereas only a logged in user can book tickets. The Login and Logout buttons should accordingly display different pages – once the user logs in, the User page is displayed; when the user clicks Logout, the Guest page is displayed again.

In this lab I have learnt how to:

- Implement conditional rendering in React applications

Step 1: Create the React Project

Created a new React project named **ticketbookingapp**:

```
npx create-react-app ticketbookingapp
cd ticketbookingapp
npm start
```

Step 2: Greeting Component (Element Variables)

Created **UserGreeting** and **GuestGreeting** components, and a **Greeting** component that conditionally renders one of them based on the **isLoggedIn** prop:

```
function UserGreeting(props) {
  return <h1>Welcome back</h1>;
}

function GuestGreeting(props) {
  return <h1>Please sign up.</h1>;
}

function Greeting(props) {
  const isLoggedIn = props.isLoggedIn;
  if (isLoggedIn) {
    return <UserGreeting />;
  }
  return <GuestGreeting />;
}
```

Code: src/Greeting.js

Step 3: LoginButton and LogoutButton Components

Created the **LoginButton** and **LogoutButton** components, each accepting an **onClick** handler as a prop:

```
function LoginButton(props) {
  return (
    <button onClick={props.onClick}>
      Login
    </button>
  );
}
```

```

}

function LogoutButton(props) {
  return (
    <button onClick={props.onClick}>
      Logout
    </button>
  );
}

```

Code: src/LoginButton.js / src/LogoutButton.js

Step 4: Flight Details and Book Ticket Pages

Created a **FlightDetails** component that the guest user can browse, and a **BookTicket** component that is only available to the logged in user:

```

const flights = [
  { flightNo: 'AI101', from: 'Delhi', to: 'Mumbai', time: '06:00 AM' },
  { flightNo: 'AI202', from: 'Mumbai', to: 'Bengaluru', time: '09:30 AM' },
  { flightNo: 'AI303', from: 'Chennai', to: 'Kolkata', time: '01:15 PM' },
];

function FlightDetails() {
  return (
    <div>
      <h2>Available Flights</h2>
      <ul>
        {flights.map((f) => (
          <li key={f.flightNo}>
            {f.flightNo}: {f.from} to {f.to} at {f.time}
          </li>
        ))}
      </ul>
    </div>
  );
}

function BookTicket() {
  return (
    <div>
      <h2>Book Your Ticket</h2>
      <ul>
        {flights.map((f) => (
          <li key={f.flightNo}>
            {f.flightNo}: {f.from} to {f.to} at {f.time}
            <button style={{marginLeft: '10px'}}>Book</button>
          </li>
        ))}
      </ul>
    </div>
  );
}

```

Code: src/FlightDetails.js / src/BookTicket.js

Step 5: App.js – Preventing Components from Rendering

In **App.js**, used the **isLoggedIn** state to decide which button and which page to render, so the Book Ticket page never renders for a guest user:

```
import { useState } from 'react';
import './App.css';

function App() {
  const [isLoggedIn, setIsLoggedIn] = useState(false);

  function handleLoginClick() {
    setIsLoggedIn(true);
  }

  function handleLogoutClick() {
    setIsLoggedIn(false);
  }

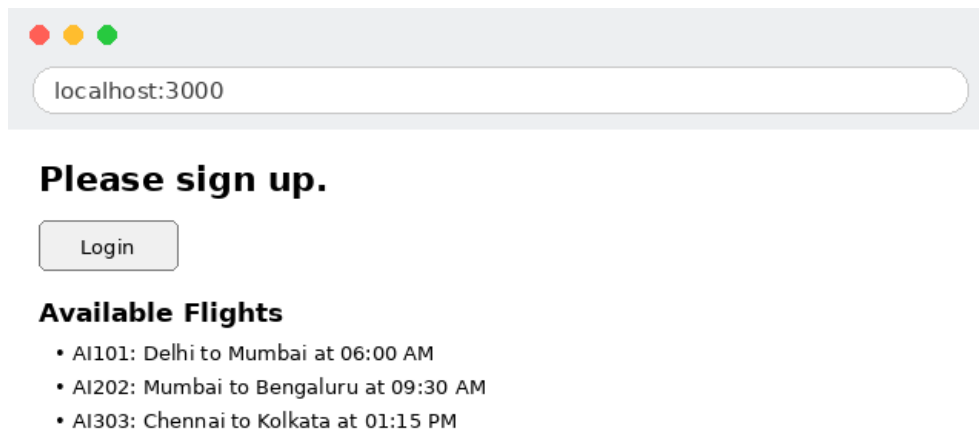
  return (
    <div>
      <Greeting isLoggedIn={isLoggedIn} />
      {isLoggedIn
        ? <LogoutButton onClick={handleLogoutClick} />
        : <LoginButton onClick={handleLoginClick} />}
      {isLoggedIn ? <BookTicket /> : <FlightDetails />}
    </div>
  );
}

export default App;
```

Code: src/App.js

Output: Guest Page (Not Logged In)

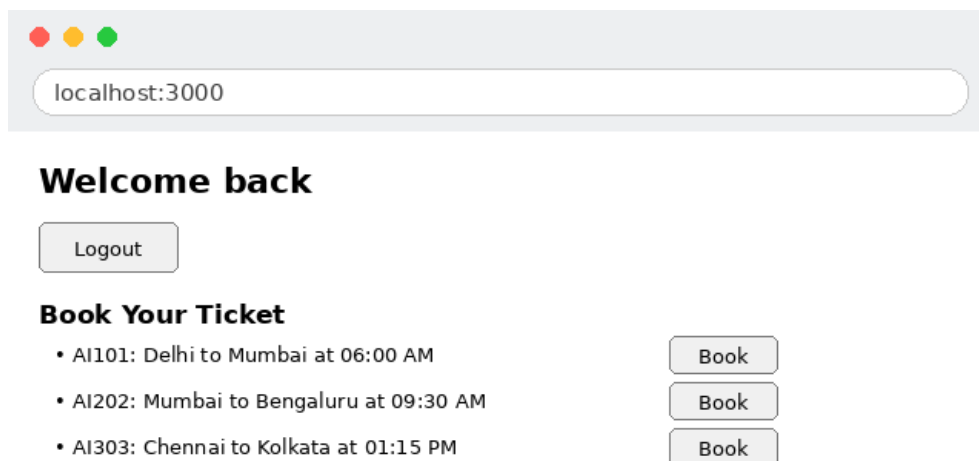
When the app loads, isLoggedIn is false, so the Guest greeting, the Login button and the flight details are displayed:



Screenshot: Guest page at localhost:3000

Output: User Page (Logged In)

After clicking Login, isLoggedIn becomes true, so the User greeting, the Logout button and the Book Ticket page are displayed:



Screenshot: User page at localhost:3000 after clicking Login

Clicking Logout on the User page sets isLoggedIn back to false and returns to the Guest page shown above.

End of Lab